

A Discrete-Time Current Controller for Permanent Magnet Synchronous Motor Drives

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Abstract—A discrete-time current controller is presented for the current loop of the permanent magnet synchronous motor drives. Firstly, the difference equations of current errors are obtained for constant reference values. Then, a dead-beat controller is designed to drive the dq -axes currents to the desired values. Discrete-time integral action with an anti-windup scheme is added to the dead-beat structure to achieve a better steady-state behaviour. In order to improve the transient properties of the controller, a current prediction scheme based on dead-beat structure is also introduced to the controller. Finally, the proposed controller is implemented through numerical simulations to illustrate the effectiveness for both transient and steady-state responses.

I. INTRODUCTION

The permanent magnet synchronous motors (PMSMs) take an important role in different application areas due to their high efficiency, power density, low electrical losses etc. comparing to other type of electric motors. Advantageous torque-speed characteristics of PMSMs make them preferable for direct-drive applications in which the torque oscillations are undesirable. However, one of the main disadvantages of PMSMs is the torque ripples.

The research on reducing the torque ripples for PMSMs can be classified into two categories, that are motor design and advanced control techniques. Although there are many machine design techniques aiming to reduce the torque ripples, they may be insufficient or inconvenient to achieve required torque ripple reduction [1]. This involves the development of the control techniques for PMSM drives to minimize the torque ripples. Since sinusoidal phase currents are required for constant torque output, the aim of the proposed controllers is to track the desired sinusoidal current references to reduce torque ripples. It is well known that one of the reasons of torque ripples is imperfections and harmonics on back emf of PMSMs. In order to reduce the effect of these undesired disturbances, back emf terms are parametrized and an adaptive controller is designed in [2]. In [3], unmodelled dynamics are considered as uncertainty, and an adaptive internal model and a reference current generation scheme are presented to ensure the robustness. Robust model reference adaptive control structure

is also proposed for PMSMs [4] to eliminate the effects of imperfect back emf signals.

Learning based control schemes are considered for the current control of PMSMs as well. A repetitive controller is designed in discrete-time for torque ripple minimization of PMSMs with harmonics on back emf in [5]. Considering parasitic torque pulsations caused by nonsinusoidal flux density distribution around the air-gap, iterative learning control structures are also applied [6], [7]. A review paper of iterative learning controller schemes is presented in [8].

In general, to reduce the torque ripples in PMSMs, the control laws are designed considering the continuous time model of PMSM. However, application and gain tuning may become a hard work for continuous time controllers since the electrical model of PMSMs may have fast dynamic response and the drives require discrete-time and switching implementation. Hence, a discrete-time robust controller [9] and a robust controller with an adaptive disturbance observer [10] are designed for the current control of PMSMs.

Another important issue for PMSM drives is the measurement delay. This issue is also attracted the attention of researchers, since these delays cause poor dynamic response of phase currents. Although the necessary computations can be performed fast, measurement delays are inevitable in switching systems. In order to overcome the effects of measurement delays, model based current prediction strategies are developed for PMSM drives [9], [11]–[13].

Among discrete-time control schemes, the dead-beat controller is studied and analysed widely due to its transient response characteristics. The necessity of the utilization of dead-beat controller comparing to PI controllers for satisfaction of high-bandwidth requirements is reported in [14]. On the other hand, the fragility of dead-beat controllers against uncertainties is also well known. A current control structure that utilizes the advantage of the dead-beat controller and combines it with a reference current prediction and an estimation procedure for back emf is presented for PMSM drive in [15]. In addition, due to the uncertainties on the motor parameters and unideal inverter operation, current error may occur especially in steady-state when model based controller and prediction are applied.

This issue is addressed in [16], and a current error-correction method is proposed to eliminate such a problem. Also, a classification and comparison of predictive current control schemes for a PMSM drive is carried out in [17] in which the sensitivity of the methods to the model parameters is analysed by means of numerical simulations.

In this paper, a discrete-time dead-beat controller in state space form is designed for the current loop of PMSM. To deal with the uncertainties, discrete-time integral action with anti-windup compensation is also added to the controller. Analysis of asymptotic convergence of the current error to zero is carried out theoretically. In order to mimic the application in the real time, the computation delay is considered and a dead-beat current prediction is also involved in the controller. The proposed controller is tested through numerical simulations by demonstrating the effects of uncertainties and delays.

The remainder of the paper is organized as follows. Section II gives the dynamic equations and defines difference equations for direct and quadrature axes currents of the PMSM. Following, general current control structure and the proposed controller are explained in detail in Section III. Section IV is devoted to present illustrative numerical simulations. Lastly, Section V concludes the paper.

II. DYNAMICS

Dynamical equations of PMSM and the error dynamics are given in this section. The following assumptions are made for the mathematical model [18].

1. Saturation is neglected.
2. Eddy currents and hysteresis losses are negligible.
3. There are no field current dynamics.
4. There is no cage on the rotor.
5. The induced EMF is sinusoidal although the other harmonics can be taken into account.

Mathematical model of the equivalent circuit of PMSM in dq -frame can be given as

$$\frac{d}{dt}i_q = \frac{1}{L_q}(V_q - R_s i_q - \omega_e \phi_d) \quad (1)$$

$$\frac{d}{dt}i_d = \frac{1}{L_d}(V_d - R_s i_d + \omega_e \phi_q) \quad (2)$$

where V_d and V_q are dq -axes applied voltages, i_d and i_q are dq -axes currents, L_d and L_q are dq -axes inductances, ϕ_d and ϕ_q are dq -axes flux linkages, R_s and ω_e are the stator resistance and electrical angular velocity, respectively. The expressions of dq -axes flux linkages and the torque produced by motor can be given by

$$\phi_d = \phi_{d0} + L_d i_d \quad (3)$$

$$\phi_q = L_q i_q \quad (4)$$

$$T_M = \frac{3}{2}P(i_q \phi_d - i_d \phi_q) \quad (5)$$

where ϕ_{d0} is the flux linkage due to the permanent magnets and P is the number of pole pairs. The mathematical expressions

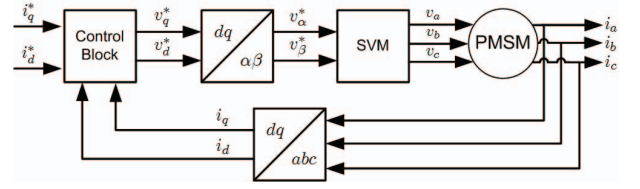


Fig. 1: The block diagram of PMSM current control system

of the mechanical part is

$$\frac{d}{dt}\theta_r = \omega_r \quad (6)$$

$$\frac{d}{dt}\omega_r = \frac{1}{J}(T_M - T_L - B\omega_r) \quad (7)$$

where θ_r and ω_r are the angular position and velocity of the rotor, J is the moment of inertia, T_L is the load torque and B is the viscous damping coefficient. The relation between electrical angular position (θ_e) and the rotor angular position is given by $\theta_e = P\theta_r$. The Park transformation that allows to obtain the dq -frame variables from abc -frame variables is defined as following

$$\begin{bmatrix} V_q \\ V_d \\ V_0 \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos(\theta_e) & \cos(\theta_e - \frac{2\pi}{3}) & \cos(\theta_e + \frac{2\pi}{3}) \\ \sin(\theta_e) & \sin(\theta_e - \frac{2\pi}{3}) & \sin(\theta_e + \frac{2\pi}{3}) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} \quad (8)$$

where V_a , V_b and V_c are the applied voltages to the stator.

Assuming that the electrical frequency of the motor is constant between two sampling instances and utilizing Euler approximation, the difference equations for (1) and (2) can be obtained as

$$I(k+1) = FI(k) + D + GV(k) \quad (9)$$

with

$$F = \begin{bmatrix} 1 - \frac{R_s T_p}{L_q} & -\frac{L_d \omega_e T_p}{L_q} \\ \frac{L_d \omega_e T_p}{L_d} & 1 - \frac{R_s T_p}{L_d} \end{bmatrix},$$

$$D = \begin{bmatrix} -\frac{\phi_{d0}}{L_q} \omega_e T_p \\ 0 \end{bmatrix}, \quad G = \begin{bmatrix} \frac{T_p}{L_q} & 0 \\ 0 & \frac{T_p}{L_d} \end{bmatrix},$$

where $k = 0, 1, 2, \dots$ is the number of the sampling instant, $I(\cdot) = [i_q(\cdot) \ i_d(\cdot)]^T$, $V(\cdot) = [v_q(\cdot) \ v_d(\cdot)]^T$ and T_p is the sampling (or PWM switching) period. Besides, the current error vector can be defined as

$$E(k) = I^* - I = \begin{bmatrix} i_q^* \\ i_d^* \end{bmatrix} - \begin{bmatrix} i_q(k) \\ i_d(k) \end{bmatrix} \quad (10)$$

where I^* denotes the reference current vector which is considered to be constant. Using (9) in (10), difference equation of the current error vector is given by

$$E(k+1) = I^* - FI(k) - D - GV(k). \quad (11)$$

The aim here is to make the current error vector to converge to zero in finite sampling instances.

III. CONTROLLER STRUCTURE

Controller design procedure is given in detail in this section. Figure 1 depicts the general block diagram of the current control system for PMSM. In order to drive current error vector to zero, it is sufficient to satisfy

$$\|E(k+1)\| < \|E(k)\|. \quad (12)$$

To satisfy this condition, the voltage vector can be defined as

$$V(k) = G^{-1}(-D - FI(k) + I^* - U(k)) \quad (13)$$

which brings (11) into the form of

$$E(k+1) = U(k) \quad (14)$$

where $U(k)$ is to be assigned. Note that, the voltage vector (control signal) given in (13) includes motor parameters, sampling period and reference current vector. Thus, it is able to cancel linear and nonlinear terms depending on the motor model, and the motor model has to be known very well to apply (13). However, not only motor parameters may vary during the operation, but there is also unmodeled dynamics such as inverter switching. These uncertainties effect the response adversely and may cause undesired dynamical behaviours. Therefore, $U(k)$ becomes a necessary signal to take care of these uncertainties. Notice that, if $U(k) = 0$, then the control signal defined in (13) is said to be the dead-beat controller which provides the fastest possible response without overshoot and steady-state error. But, fragility of this type of controller against uncertainties is also known well. To overcome this problem, discrete integral action is added into the control signal exploiting $U(k)$. To do so, $U(k)$ is designed

$$U(k) = -K_I \sum_{i=1}^k E(i) \quad (15)$$

where $K_I \in \mathbb{R}^{2 \times 2}$ is symmetric gain matrix. Utilizing (15) in (14) and rearranging series yield

$$E(k+1) = -K_I(-K_I + I_2)^{(k-1)} E(1) \quad (16)$$

where I_2 is 2×2 identity matrix. The gain matrix K_I has to be selected to satisfy (12) in order to provide asymptotic convergence. Note that, if K_I has any nonpositive eigenvalue, then the condition (12) is not satisfied and therefore the error vector becomes unbounded as $k \rightarrow \infty$. Similarly, if $\lambda_{max}(K_I) > 2$ where λ_{max} is the maximum eigenvalue of K_I , asymptotic convergence can not be provided either. On the other hand, it is clear that if $0 < \lambda_{1,2}(K_I) < 2$ is satisfied, $(-K_I + I_2)^{(k-1)}$ decreases as k increases, and therefore $E(k)$ also decreases. Hence, $0 < \lambda_{1,2}(K_I) < 2$ has to be satisfied to obtain asymptotic convergence.

In addition to (16), conditional integration method, [19], is added to the structure of the controller, in order to reduce the effect of the integral action to transient response. In this manner, if the inverter is in saturation, then the integral action is cancelled for the corresponding sampling instant.

The controller (13) with (15) is supposed to be implemented at k th time instant. However, it is not possible in a practical application because of the delay on the measurements and the mathematical computations. Hence, it is worth introducing a current prediction procedure into the controller to

TABLE I: PMSM parameters

Symbol	Description	Value	Unit
	Rated power	1350	W
T_n	Rated torque	7.62	Nm
I_n	Rated current	60	A
ω_n	Rated speed	1700	rpm
P	Number of pole pairs	6	
ϕ_{d0}	Flux linkage due to the permanent magnets	0.01	Wb
L_d	d -axis inductance	24.7	μ H
L_q	q -axis inductance	24.8	μ H
R_s	Stator resistance	7	m Ω

achieve a better current response. Following the idea given in [12], a current prediction algorithm is utilized in this study based on the dead-beat structure. To do so, the continuous time model for dq -axis currents given in (1) and (2) is considered to be discretized for sampling time T_d that is the delay on the measurement. Hence, the discretized model is given by,

$$\hat{I}(k+1) = F_p I(k) + D_p + G_p V(k) \quad (17)$$

with

$$F_p = \begin{bmatrix} 1 - \frac{R_s}{L_q} T_d & -\frac{L_d}{L_q} \omega_e T_d \\ \frac{L_q}{L_d} \omega_e T_d & 1 - \frac{R_s}{L_d} T_d \end{bmatrix},$$

$$D = \begin{bmatrix} -\frac{\phi_{d0}}{L_q} \omega_e T_d \\ 0 \end{bmatrix}, \quad G = \begin{bmatrix} \frac{T_d}{L_q} & 0 \\ 0 & \frac{T_d}{L_d} \end{bmatrix},$$

where $\hat{I}(k+1)$ is the predicted current vector for the beginning of $(k+1)$ th PWM period and $V(k)$ is reference voltage vector applied in k th PWM period. All the other model parameters and states are as defined in (9). In a practical application using the dead-beat current prediction, the current measurement should be performed at the beginning of the PWM period to avoid the switching ripple noise on the currents. Besides, if PWM duties are symmetric in a switching period, then the measurement can also be performed at the middle of the PWM period which is the case in this paper. Hence, the predicted current vector $\hat{I}(k+1)$ is obtained via (17) for the sampling period $T_d = T_p/2$.

IV. NUMERICAL SIMULATIONS

The proposed current loop controller is implemented through numerical simulations to show its effectiveness. PMSM model parameters considered in this paper are given in Table I. Mechanical part of the motor is neglected and the motor speed is assumed to be constant throughout the simulations. It is considered that the current measurements are performed in the middle of the corresponding PWM period to be used in the controller. Following that, the controller gives the reference voltage values for the next period. Thus, there is a half PWM period delay between the instant and measured values of currents. DC bus voltage is set to 21V and dead-time of the inverter is neglected in simulations. Numerical simulations are implemented via MATLAB. PWM resolution is set to 50ns, and simulation data are recorded at the inverter switching instants. Standard symmetric space vector PWM method is utilized throughout the numerical simulations.

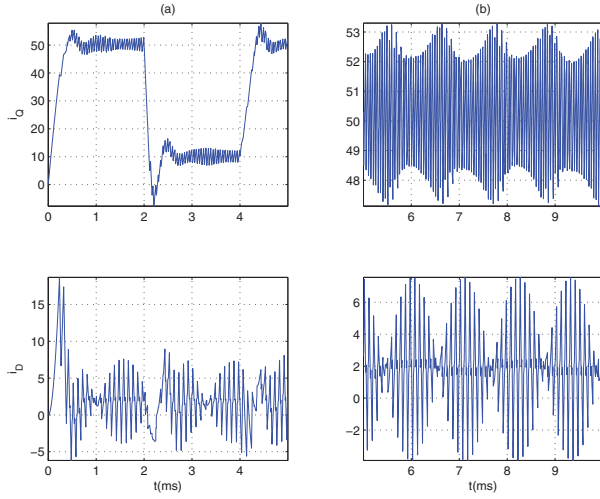


Fig. 2: Simulation result for case-1.

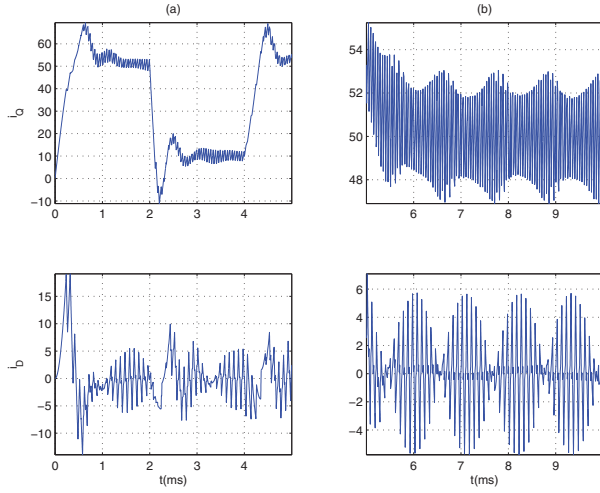


Fig. 3: Simulation result for case-2.

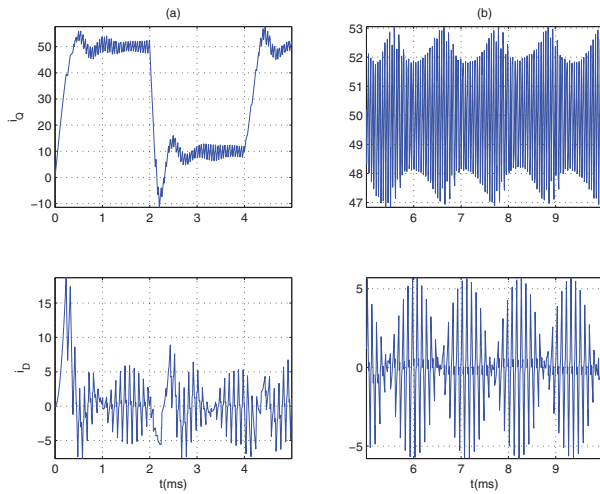


Fig. 4: Simulation result for case-3.

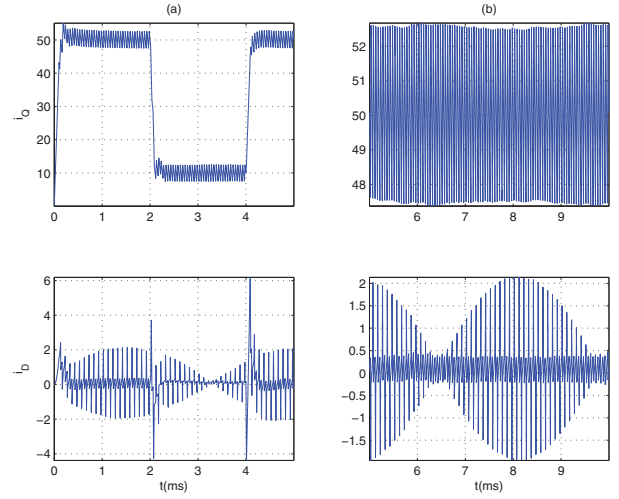


Fig. 5: Simulation result for case-4, $\omega_R = 500rpm$ and $k_i = 0.2$.

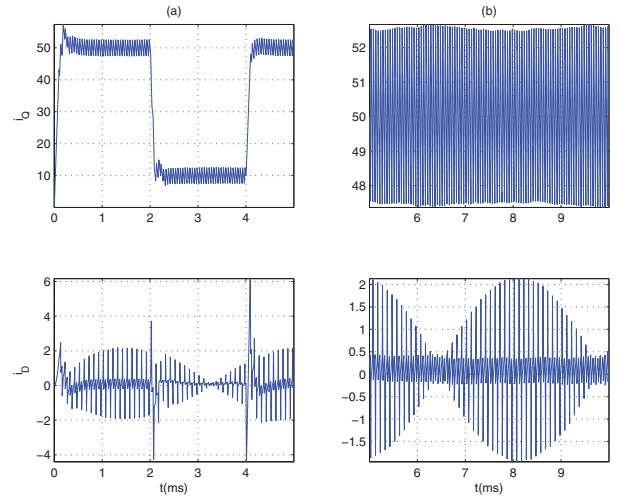


Fig. 6: Simulation result for case-4, $\omega_R = 500rpm$ and $k_i = 0.5$.

Various simulation studies are performed to illustrate the improvement and the performance of the proposed controller. These studies are classified into three cases and summarized as following,

- Case-1: Only the dead-beat controller is implemented.
- Case-2: The integral action without an anti-windup operation is added to the dead-beat controller to eliminate steady-state error.
- Case-3: The integral action with conditional anti-windup scheme is added to the dead-beat controller to achieve a better current response.
- Case-4: The proposed controller with current prediction is implemented to improve the transient response. The simulation studies belonging to this case are performed for different rotor speeds and integral gains to show the effectiveness of the proposed controller structure.

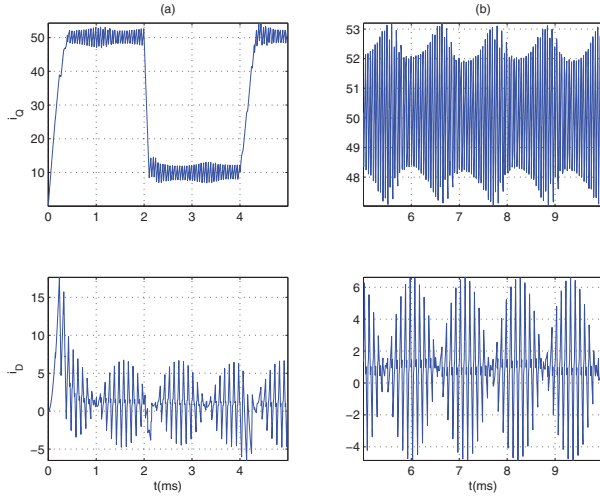


Fig. 7: Simulation result for case-4, $\omega_R = 1500rpm$ and $k_i = 0.2$.

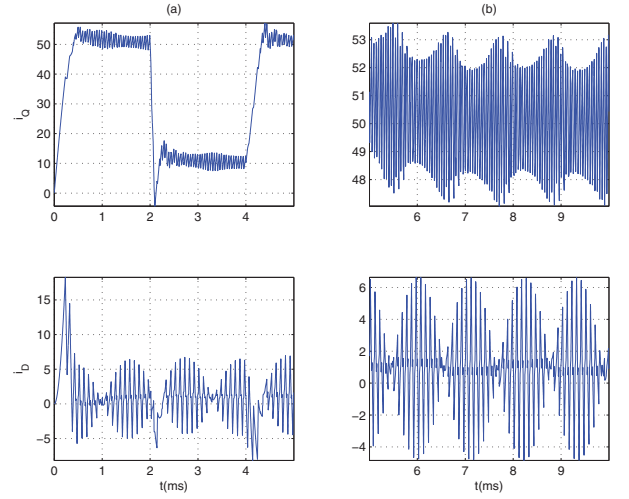


Fig. 9: Simulation result for case-5, $\omega_R = 1500rpm$, $k_p = 0.4$ and $k_i = 0.5$.

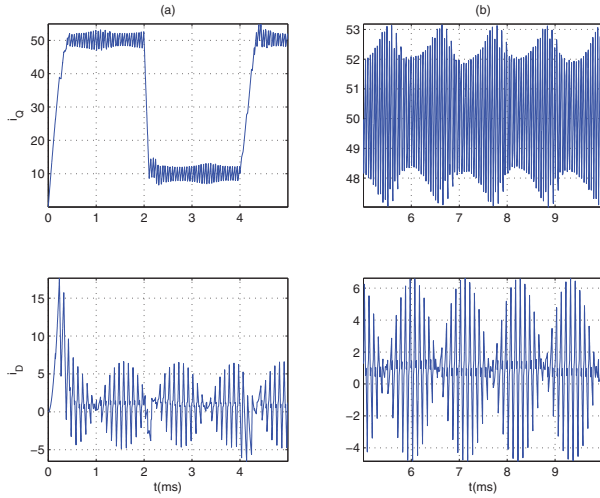


Fig. 8: Simulation result for case-4, $\omega_R = 1500rpm$ and $k_i = 0.5$.

Case-5: Conventional PI controller is implemented in order to compare the performance of the proposed controller to the PI controller.

The integral gain matrix is adjusted as

$$K_I = \begin{bmatrix} k_i & 0 \\ 0 & k_i \end{bmatrix} \quad (18)$$

with $k_i \in \mathbb{R}$ for the simulations in which integral action is utilized. The parameters for each simulation are given in Table II. All the initial conditions are set to zero and numerical simulations are run for 10ms. Reference d -axis current is fixed to zero while reference q -axis current is set to 50A at the beginning, 10A between 2 – 4ms and 50A after 4ms.

Table II also presents maximum ripple (Δi) and average error ($e_{i,avg}$) regarding to dq -axis currents in steady-state. These values are computed using the data for the last 2ms of the simulations in order to examine the steady-state response.

TABLE II: Simulation parameters and results

Case	ω_R	k_i	Δi_q	$e_{iq,avg}$	Δi_d	$e_{id,avg}$
#1	1500	-	6.175	0.253	11.475	1.872
#2	1500	0.2	6.101	0.001	11.441	0.007
#3	1500	0.2	6.109	0.003	11.461	0.006
#4	500	0.2	5.272	0.013	4.082	0.112
#4	500	0.5	5.296	0.01	4.082	0.113
#4	1500	0.2	6.151	0.12	11.44	0.931
#4	1500	0.5	6.156	0.125	11.46	0.936
#5	1500	0.2	6.083	0.125	11.454	0.94

Results for average values are quite different for each case of simulation. Comparing the simulation cases for the same rotor speed and the value of the integral gain, we can see that the best result in the steady-state is obtained with dead-beat+integral controllers. This shows the robustness of the proposed dead-beat+integral controller to the model uncertainties. On the other hand, higher values of the average error is obtained for the controller with current prediction comparing to pure dead-beat+integral controller. This shows the effects of the uncertainties (especially PWM switching) on the fragile dead-beat procedure. Besides, the maximum ripple value is larger in the simulations with 1500rpm rotor speed comparing to the simulations with 500rpm rotor speed. This is an expected result since the ripple on the dq -axes currents increase as the motor speed increases.

Figures 2-9 depict the results for the numerical simulations. Each plot is presented in two distinguished parts for 0 – 5ms and 5 – 10ms to analyse transient and steady-state responses separately. Note that, a relatively large current ripple with high frequency has occurred in the results due to the PWM switching, and the small value of the phase inductances. On the other hand, space vector PWM also causes fluctuations on the dq -axes currents that are clear in the figures especially for the simulations with 1500rpm rotor speed. The result for dead-beat and dead-beat+integral controllers are given in Fig. 2-4. It can be observed that the response of the dead-beat controller is not so satisfactory since overshoots and the

steady-state error occur in the results. Overshoots also occur when dead-beat+integral controller is implemented. However, integral action carries out a good result on the steady-state response as discussed previously. In addition, comparing Fig. 3 and Fig. 4, one can observe the effect of the anti-windup operation on the transient response. The reason for the overshoots in transient response is the delay in current measurements. It is clear from Fig. 5-8 that the transient response can be improved when the current prediction is introduced to the dead-beat+integral controller. The results for 500rpm rotor speed are presented in Fig. 5 and Fig. 6 for different integral gains, namely $k_i = 0.1$ and $k_i = 0.5$. Comparing these results, it can be observed that an increase on the integral gain may cause more overshoot while it has almost no effect on the steady-state. The same result is also valid for 1500rpm rotor speed which are presented in Fig. 7 and Fig. 8 for different integral gains. A comparison between Fig. 5 and Fig. 7 (Fig. 6 and Fig. 8) shows that the transient response properties do not change much with the rotor speed. However, the shape of the currents in steady-state changes with the rotor speed which is caused by the modulation.

Finally a comparison between the proposed controller and conventional PI controller is performed through simulations. Fig. 9 depicts the response of the PI controller. The proportional gain matrix is selected as a diagonal matrix that is similar to the integral gain matrix given in (18) in which k_i is replaced by k_p . To obtain a similar transient response, proportional gain is set to $k_p = 0.4$ for the simulation while the integral gain matrix is hold the same structure as in the simulations of the proposed controller. Conditional anti-windup and current prediction schemes are also added to the PI controller. The result of the PI controller for steady-state is presented in Table II. It can be concluded that there is almost no difference between the steady-state responses of the PI controller and the proposed controller. However, the same conclusion can not be done for transient performances. A comparison between Fig. 7 and Fig. 9 shows that the transient response of the proposed controller is better than the conventional PI controller.

V. CONCLUSION

A discrete-time current control structure has been presented for PMSMs. The proposed controller consists of the dead-beat controller and the integral action with anti-windup compensation in order to reduce the steady-state error. A current prediction scheme based on dead-beat structure is also introduced to the controller to improve the transient performance. Different numerical simulation studies have been performed to show the contribution of both integral action and current prediction, and to test the designed controller under different conditions. Because of the fragile structure of the utilized prediction scheme, undesired steady-state performance has been observed in the simulation results.

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